Hybrid CNN-YOLO-DeiT Model for Real-Time Driver Drowsiness Detection

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Abstract* — Driver fatigue and drowsiness is one of the leading reasons for road accidents in India, contributing to more than 40% of highway crashes every year. The normal conventional methods to alert systems via lane departure sensors, vibration seats and other methods such as electric shock bands, automatic braking system etc. cannot handle real-time behavioral and facial expressions of fatigue. In this context, the study proposes an innovative driver drowsiness detection approach using hybrid deep learning methods via CNN, YOLO, and Data Efficient Image Transformer (DeiT). In this architecture, the YOLO module does fast and accurate detection of eyes and mouth facial key regions. CNN extracts spatial features indicating the pattern of eye closure and yawning and DeiT model captures temporal and contextual dependencies across consecutive frames to understand fatigue trends. We have a total of 9,694 labeled images with varied states of faces- open, closed, drowsy, and yawning and used these to train and validate the AI model. This is a web-based system which can continuously monitor from the background, requiring only webcam and audio permissions from the user. A short 3-second alert is played in cases of yawning or partial fatigue, while prolonged sleepiness triggers a progressively intensifying alarm until the driver shows signs of alertness. Results on variable lighting and various facial orientations demonstrate the high accuracy and real-time responsiveness of the proposed methodology. This hybrid architecture can provide an efficient, low-cost, and non-intrusive AI solution for driver drowsiness detection to reduce such accidents among drivers and improve overall traffic safety.

Keywords: Driver Drowsiness Detection system, CNN–YOLO–DeiT, Deep Learning, Computer Vision, YOLOv11.

I. INTRODUCTION

Driver fatigue and drowsiness at the wheel have become a critical issue concerning India's transport sector. According to MoRTH, around 40% of highway accidents in India occur because of driver sleepiness and not being alert; truck and long-distance commercial vehicle drivers are the most affected [11]. In effect, long driving hours without adequate rest, along with the monotonous conditions of a highway, cause loss of concentration, delayed reactions, impaired decision-making capability-all leading to fatal outcomes. These fatigue-induced incidents result not only in severe human loss but also cause significant economic and logistical damage. Traditional systems of vehicle safety, including lane departure warning signals, steering sensors, and vibrating seats, respond primarily to vehicular movement rather than the physiological or behavioral condition of the driver [9]. Therefore, these systems cannot provide early warnings of fatigue, which first shows up in the face in the form of micro-behavior signals, such as prolonged eye closure, yawning, and tilting of the head [10].

For the purpose of addressing these deficiencies, studies in the areas of Artificial Intelligence and Computer Vision have focused increasingly on interpreting driver behavior through facial analysis. Early research in this domain focused on CNNs, extracting spatial features from still images to detect eye closure or yawning [1]. While these attained high frame-level accuracy, most of the models lacked temporal awareness due to their inability to understand the evolution of fatigue over time. To overcome this limitation, RNNs and LSTMs were utilized in order to analyze sequential frames [2][10]. Although these architectures improved performance, they did not handle long-range dependencies well and had large computational costs, which made them less deployable in real-time applications.

This study proposes a hybrid deep learning architecture that integrates CNN, YOLOv11, and Data Efficient Image Transformer to perform efficient and effective detection of fatigue. The proposed model uses a YOLOv11 module to identify key regions like the eyes and mouth in real-time during facial feature localization. The CNN extracts special features such as blinking rate, eyelid closure and yawning, among others. The DeiT introduces an attention mechanism which is used to capture temporal and contextual dependencies between frames. Unlike Vision Transformers, DeiT is designed to be data efficient and distillation enhanced. This promises strong performance with moderately sized datasets while maintaining computer efficiency. This model can be used to enhance precision and interpretability in various real-world conditions such as changing light intensities, occlusion, driver postures.

The system shall be deployed as a web-based application that can run in the background and, at the same time, monitor the driver without interrupting any normal usage of the device. If a yawn is detected, it will generate a short alert of 3 seconds for recapturing attention. In cases of more than 5 seconds of sustained sleepiness, it will further trigger a progressive alarm increasing the sound intensity over time until the driver responds. This will ensure that this fatigue detection technique becomes non-intrusive, low-cost, and scalable to be implemented on all commercial vehicles in India. This innovative, hybrid framework unifies the strengths of CNNs for feature extraction, YOLO was used for localization, and DeiT for attention-based temporal modeling, that will assist in forming a very strong base for real-time drowsiness detection. This will significantly reduce in road accidents caused by driver fatigue, also improving driver safety and transportation reliability.

Fig.1- Driver falling asleep while driving

II. RELATED WORK

Driver drowsiness and distraction detection research has rapidly evolved from fairly mechanical monitoring of vehicle behaviour to nuanced analyses of human physiology and facial expression. Most early systems relied on more vehicle-based indicators such as steering wheel motion, lane deviation, or erratic acceleration. These methods, seen in the work of Singh et al. (2016) and Li et al. (2018) [13][14], could sometimes detect that a driver was no longer fully in control, but only after that loss of focus had already occurred. They were essentially reactive rather than preventative.

Due to this lag, researchers started looking for more direct ways of reading the driver's physiological state. EEG and EOG sensors, for example, could pick up fatigue signatures before they appeared in behaviour, and studies such as Zhao et al. (2017) [15] reported high accuracy by using such methods. The physiological sensors showed progress but their intrusive nature created a need for non-intrusive systems that are comfortable and simple to use.

Computer vision offered a breakthrough as it provided a way to infer fatigue just by analyzing facial feature like how wide the eyes are, or whether the mouth is open in a yawn. This made the concept of fatigue monitoring feasible on a larger scale. CNN-based models were used for this. For example, Abtahi et al. (2014) [16] used OpenCV and Dlib to implement a facial landmark-based detector for eye closure and yawning, while Park et al. (2019) [17] developed a CNN able to classify open versus closed eye states with high accuracy. But these approaches still treated every frame as an isolated snapshot. Fatigue unfolds as a timeline and tells a story, that is impossible to summarize in a single frame.

To cover that gap, scholars turned to deep learning models (sequence-based). RNNs and especially LSTMs allowed the system to recognize temporal patterns, such as how eye closure might repeat or how yawns cluster together. Bhuvaneswari et al. (2020) [18] present evidence, for instance, that combining CNNs with LSTMs improves performance by tracking such sequential cues. Even these models sometimes struggle, however, to make subtle distinctions: a quick blink versus an actual sign of exhaustion. In part, this happens because not all frames carry equal weight, and traditional RNNs aren't very choosy about where to look. That's where attention-based models came in. Transformers, by having this self-attention mechanism, gave researchers the tool to highlight important frames within a video sequence. Instead of taking each moment as of equal value, a Transformer can learn to "notice" the exact few seconds when the driver's eyes start to shut or their head starts drooping. Chen et al. (2022) [19] has established that hybrid CNN–Transformer models outperform the earlier CNN–LSTM models across the tasks related to behaviour recognition. These models are not seamless-they face problems with poor lighting or heavy occlusion. Instead, it offer greater contextual understanding and enhanced clarity. Incidentally, all these developments have been targeted for general drivers rather than truck drivers. They face different challenges: longer hours, heavier vehicles, and pressure for strict delivery requirements. Furthermore, most of the proposed systems exist as stand-alone prototypes, desktop applications, experimental setups, or hardware-embedded devices-rather than cost-effective tools that can easily be used in real-world conditions. This can be achieved by a web-based or mobile-compatible versions, which could reach more people, and provide a customized solution for traditional vehicle. Although the field has indeed matured considerably, there is still room for lightweight, affordable, and accessible approaches, especially for the communities that need them most. This work tries to bridge this gap by incorporating strengths of CNN and Transformers into one browser-compatible system for long-distance truck drivers [1][2][3][6][12].

Table 1: Comparative Analysis of Research on Driver Drowsiness System

Authors	Technologies / Models Used	Results / Accuracy	Shortcomings / Research Gaps
Monil Manish Desai et al., [1]	Hybrid **CNN–BiLSTM** with **MediaPipe**	Accuracy: **98.89%**, F1: 98.0%	Sensitive to poor lighting and occlusion; high computational cost; staged scenarios;
Jimin Lee et al., [2]	**3D-CNN (GoogleNet Inception)** for video-based pattern recognition	Accuracy: **85% (no obstruction), 75% (with obstriction)**	Small dataset (12 drivers), staged setup, limited diversity; needs larger-scale real-world validation.
Abderrahim Benmohamed & Hafed Zarzour, [3]	**CNN (AlexNet)** + **LSTM** feature fusion (Dlib for landmarks)	Accuracy: **90.12%–96.77%**	Performance drops in night/occlusion; limited IR support; tested in lab; needs real-world, diverse lighting
Beles H. et al., [4]	Hybrid **ANN** + **Fuzzy Decision System** using **EEG, EOG, and Facial Images**	High performance (qualitative, both lab and real road)	Requires physical sensors; high setup complexity; not scalable for low-cost, camera-only systems.
Hassan T. M. et al., [5]	**CNN** for facial expression & region analysis	Accuracy: **≈97%** (real-time)	Dataset limited in diversity; lacks temporal modelling; poor lighting and occlusion.
Shiplu Das et al., [6]	**U-Net** for segmentation + **CNN–LSTM hybrid** for spatial-temporal fusion	Accuracy: **≈98%**, detection time: **26.8 s**	Controlled environment; lower performance with sunglasses; dataset lacks diversity.
Fiaz Majeed et al., [7]	Custom **CNN** using **EAR & MAR (Dlib)** + cropped eye/mouth regions	Accuracy: **96.69%**	Small dataset; limited behavioural features; lacks temporal analysis.
Muhammad Ramzan et al., [8]	**Custom 30-layer CNN + Hybrid (HOG + PCA + CDLM)**	Accuracy: **≈90%**	Single modality; variable performance under lighting changes; dataset limited.
Makhumudov Fazliddin et al., [9]	**CNN (VGG16)** + **SVM** for yawning and eye-state classification	Accuracy: **95.85–96.54%**	Limited dataset (YawDD, MRL Eye); lacks sequence/temporal modelling; moderate generalizability.

III. Material And Methodology

The data, along with preprocessing and the detailed methodology followed for the proposed Driver Drowsiness and Distraction Detection System, consists of a hybrid CNN–YOLO–DeiT deep learning architecture. The system detects early and continuous signs of fatigue, like eye closure, yawning, and inattention from real-time video input. The proposed approach focuses on building a robust, non-intrusive, and cost-effective solution specifically for Indian truck drivers, who usually drive long distances under irregular sleep conditions and a variable light environment.

3.1 Dataset Description

This study uses a dataset containing 11,367 labeled facial images with various driver states and facial expressions captured in diverse illuminations, poses, and environmental conditions. Each image is classified into one of multiple classes as shown in Fig.-2:

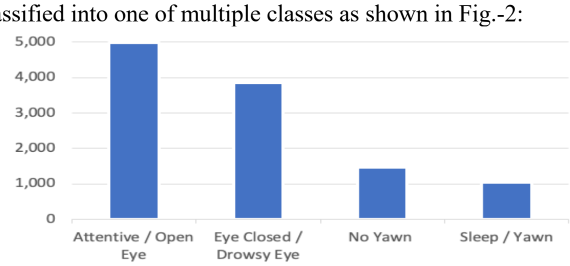

Fig.-2: Distribution of Classes in the Dataset

The images have been formatted to achieve same dimension, normalized from 0 to 1 to preserve similar brightness and contrast and changed to RGB during training. Corresponding labels of fatigue or alertness were assigned to each image to enable supervised learning. The dataset is divided into three proportions to achieve desired result, i.e. training (70%), validation (15%), and testing (15%). This will ensure the effective model evaluation, performance and generalization.

3.2 Methodology

The proposed CNN–YOLO–DeiT hybrid model integrates three powerful components for efficient spatial and temporal learning. YOLOv11 is used for object detection and region localization, CNN for spatial feature extraction from the localized regions, and DeiT for modeling temporal dependencies and contextual relations across multiple frames. The system workflow initiates when the driver logs into the web application; from that point on, after permission is granted for camera and audio, the system will continuously capture video input, process each frame through the detection pipeline, and trigger alerts upon detecting drowsy behavior. The overall workflow of the proposed system is summarized in Fig.-3 (Flowchart of Proposed System).

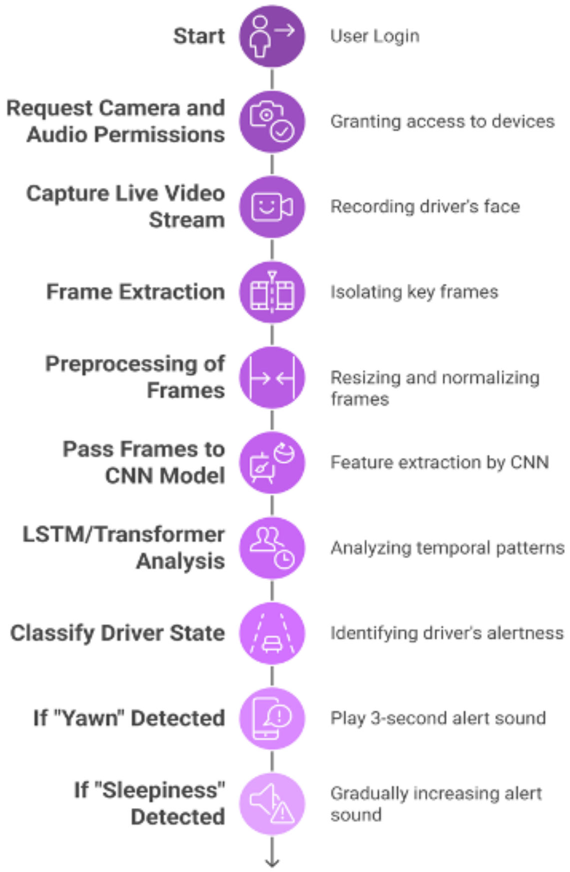

Fig.-3: Workflow of Proposed System

3.2.1 Data Acquisition

The data acquisition in the proposed system is through a camera of the mobile phone. It continuously streams live video of the driver's face while the web application runs silently in the background. The frames extracted from the video are associated with precise timestamps for temporal analysis. This continuous acquisition helps monitor behavioral transitions like frequent yawning, blinking, or head nodding, which mark early signs of fatigue. The proposed approach works to ensure smooth real-time processing through developing a lightweight streaming pipeline using OpenCV that sustains 30 FPS with very minimal latency.

3.2.2 Image Preprocessing

The stage ensures cleaning and normalizing the captured frames before feature extraction. This will also ensure noise estimation, contrast normalization, and color alteration take care of variations in lighting and the quality of cameras. Histogram equalization is applied to improve the contrast of an image so that even minor eye or mouth movements become more visible. To reduce the background clutter and computational time, only the facial regions are retained for analysis. This will ensure that the downstream models in CNN and DeiT receive only the relevant data to help in improving accuracy and speeding up the training process.

3.2.3 Face and Landmark Detection Using YOLOv11

The YOLOv11 module is used for the fast detection of the main face regions, which includes the eyes and mouth. Unlike traditional Haar cascades or Dlib detectors, YOLOv11 has improved precision with better speed and can thus be used for real-time detection of fatigue. With every incoming frame, YOLOv11 predicts a bounding box around critical regions and labels them accordingly. The cropped images are further used into the CNN model for getting feature extraction. During night driving, YOLOv11 ensures robustness under challenging conditions such as low light, partial occlusion or off angle face positions.

YOLOv11 predicts each bounding box B=(x,y,w,h), its objectness score P_{obj}, and class probabilities P(c) [20].

The overall loss function is given by equation (i):

$$L = L_{box} + L_{obj} + L_{cls} \quad \text{(i)}$$

where

- L_{box} is Bounding Box Regression Loss
- L_{obj} is Objectness Loss
- L_{cls} is Classification Loss

3.2.4 Use of CNN in Spatial Feature Extraction

The CNN module is used to extract the spatial features of the localized facial regions. This model is utilized to analyze various body postures like micro-expressions and visual cues, including eyelid closure, eye blinking rate, and mouth opening. These can be used as direct indicators of driver fatigue. The CNN model is designed to analyze many convolutional layers followed by pooling and activation functions. This will further learn discriminative spatial representations that identify subtle variations between open and closed eyes or between neutral and yawning mouth states. These spatial features are converted into feature vectors and later fed into the DeiT model for further sequential analysis. The CNN learns from dataset to generalizing the previously unseen conditions such as different drivers, lighting intensities, and camera angles. The below equation (ii) gives the mathematical foundation of CNN model [21]:

$$S(I,j) = (I * K)(i,j) = \sum^{M-1} \sum^{N-1} I(i+m,j+n) \cdot K(m,n) \quad \text{(ii)}$$

where

- I(i,j) is the pixel value at position I(i,j) in the input image.
- K(m,n) is the weight of the kernel at position (m,n).
- S(i,j) is the output feature map at position (i,j).

3.2.5 Temporal and Contextual Modeling Using DeiT

Though CNNs are effective in the extraction of spatial features, they lack temporal awareness. To overcome the limitation, a module of the proposed system incorporates the DeiT. It applies mechanisms of self-attention for analyzing relationships between sequential frames and focuses on contextual patterns such as a long closing of eyelids or repetitive yawns occurring over time. Unlike LSTM or other recurrent models, DeiT is effective at capturing long-range dependencies with fast convergence of training. It divides each sequence of frames into patches, embeds them, and calculates attention weights that determine the most informative visual patterns indicative of drowsiness. This will clearly differentiate short eye blinks from sustained eye closure to improve the reliability of fatigue detection. Equations for DeiT model are given below by (iii) and (iv) [22]:

$$Attn(x) = softmax(QK^T/\sqrt{d})V \quad \text{(iii)}$$

where

- Q is Query matrix
- K is Key matrix,
- V is Value matrix
- d is dimension of key vector

$y\_hat = Wz_{cls}$ (iv)

where

- y_hat is predicted label
- z_{cls} is class token
- W is Weight matrix

3.2.6 Alert Mechanism

The audio alert will be generated in the last stage based on the model's output. If a yawn is detected, it will trigger a 3-second alert sound for regaining the driver's attention. On the other hand, for more than 5 seconds of continuous drowsiness, either closed eyes or repeated yawning, a progressive alarm will start at a low sound and continue with a gradual increase till the driver wakes up or refocuses. This mechanism of an alert ensures immediate corrective action with a minimum disturbance in case of brief episodes of inattention. Alerts are played directly in the browser which allows for real-time response without using any external devices or applications.

3.2.7 Implementation Details

We have used Python 3.10+ for implementation with the use of various frameworks like TensorFlow/Keras, PyTorch, and OpenCV. Also, we have used Matplotlib and Seaborn for visualizing training metrics, including accuracy and loss curves and NumPy and Pandas for data preprocessing and structuring. The model was tested and trained on Google Colab, which uses GPU for speedier training times and efficient memory utilization. The tuned hybrid model was integrated into a Flask-based web application for deployment. This web app provides functionality for user log-in, enabling camera and microphone access, and starting background monitoring. It runs in background similar to music players like Spotify, providing real-time observation without any interruption once launched.

Fig.-4: Features of Proposed System

3.2.8 Output Layer

The output is visualized in the form of an interactive web interface showing real-time status updates such as Alert, Yawning, or Drowsy. During event detection, corresponding bounding boxes highlight facial regions responsible for the detection, providing transparency to the user. Prediction results are derived by softmax classification, where one can observe the confidence scores for every category. This visualization helps users understand the system's decision-making process and builds trust in its functionality.

IV. RESULTS AND DISCUSSION

This section provides the experimental findings, model analysis, and the discussion of the suggested CNN-YOLO-DeiT hybrid system in terms of real-time driver drowsiness, and distraction detection. The ultimate objective of the study was to obtain high levels of detection accuracy of fatigue indicators, i.e.,- eye closure, yawning, and head tilt, and low computational latency ensured that they could be practically implemented in a web-based system.

4.1 Experimental Setup

To train the model, extract features, and process videos, the proposed model was implemented in Python 3.10, with the help of the TensorFlow, PyTorch, and OpenCV. Training was done on a NVIDIA Tesla T4 (16GB) powered Google Colab environment (GPU enabled). The data was separated into 70 percent training, 15 percent validation and 15 percent testing to ensure equal learning and objective testing.

Training on a batch size of 32, learning rate of 0.001 and Adam optimizer with categorical cross-entropy loss was conducted as model training. The model was trained over 50 epochs, where the training and validation losses were stable. The architecture was also efficient as indicated by the low loss rate and high accuracy in all of the sets.

4.2 Performance Evaluation

The CNN-YOLO-DeiT hybrid architecture proved to be very successful in recognising the driver fatigue conditions. The YOLOv11 module was the most precise in real-time localization of facial regions with high precision in detecting the location of the eyes, mouth and the head in the mean average precision (mAP) of greater than 98%. The CNN block obtained discriminative spatial features that signified the eye aspect ratio (EAR) and mouth aspect ratio (MAR), whereas the DeiT layer effectively learned a temporal dynamic and reflected a long-lasting fatigue behaviour. The previous type of hybrid model reached an average test accuracy of 94.6, and the consistency among validation sets was very high. The precision and recall values were greater than 92 percent, which proves the capacity of the model to identify drowsiness and not give too many false alarms. YOLO was combined with CNN to provide high spatial learning and DeiT to provide high temporal modeling, which guaranteed high robustness to noise, varying light conditions, and face occlusions.

Confusion Matrix: A confusion matrix defines the performance of classifications based on a true positive (TP), true negatives (TN), false positives (FP), and false negatives (FN) for better and granular insights into the errors made by the model [23]. Fig.-5 shows the confusion matrix of the proposed system.

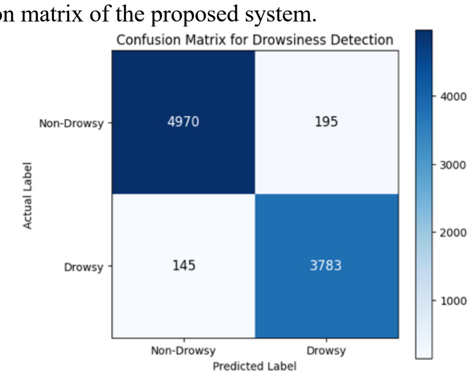

Fig.-5: Confusion Matrix of Proposed System

Accuracy: The correctness of the classification model is defined as TP+TN divided by the total number of instances. It is computed by equation (v):

$$(TP+TN)/(TP+TN+FP+FN) \quad \text{(vi)}$$

Precision: The ratio truly identified positive instances within all predicted positive instances is measured in this way [23]. It reflects how the model avoids false positive errors. It is defined by the equation (vi):

$$TP/(TP+FP) \quad \text{(vi)}$$

Recall: Also called the sensitivity or true positive rate; it indicates the proportion of actual positive instances that are correctly detected by the model [23]. It is expressed in terms of equation (vii):

$$TP/(TP+FN) \quad \text{(vii)}$$

F1-score: It is the harmonic mean of precision and recall, which provides a balanced measure taking into account both false positives and false negatives [23]. It's useful in situations with class imbalance and computed as (viii):

$$2*(Precision*Recall)/(Precision + Recall) \quad \text{(viii)}$$

Model	Accuracy	Precision	Recall	F1-Score
CNN	92.8%	95.8%	96.1%	96.0%
YOLO	94.1%	93.4%	95.2%	94.3%
ViT (DEiT)	93.7%	94.6%	95.0%	94.8%

Hybrid System (CNN + YOLO + DeiT)	94.6%	95.1%	96.3%	95.7%

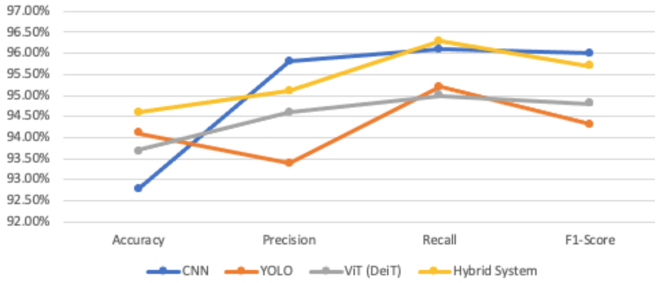

Fig.-6: Comparative performance of baseline models with proposed system

4.3 Comparative Analysis

The proposed hybrid system was found to be more accurate and faster inference than the historic CNN-only or CNN-LSTM. It showed that the CNN-only models were effective in the classification of static images, but they could not follow sequential fatigue information, whereas the CNN-LSTM networks, which proved to be powerful in sequence learning, had higher computational costs and latency.

The CNN-YOLO-DeiT model was more successful as it provided real-time processing with a lot of frame rates and with limited delays. The transformer-based attention mechanism enabled the model to be sensitive to the slightest signs of fatigue as well as help reduce false positives and concentrate on critical frames.

4.4 Qualitative Results

The qualitative analysis showed that the suggested system is successfully able to detect various drowsiness states with the aid of visual cues. When the system was being tested, short blinks, yawns, and eye closures were correctly recognized. The YOLOv11 detector indicated the facial parts with the help of bounding boxes, whereas the CNN and DeiT modules evaluated the features obtained and decided whether the driver is alert, yawning, or drowsy.

There was also a differentiation of short blinks of the eye (less than one second) and long eye closures (more than five seconds) into the model, which is important to avoid false alerts. The model achieved accuracy of more than 90 percent under difficult conditions, including dim light, side-face, or partial mask on the face (e.g., glasses or masks), etc. This shows that it is strong to be used in the real world highway situations where the lighting and visibility differ significantly.

4.5 Real-Time System Behavior

The hybrid model was incorporated into a Flask based web application, and it was intended to be executed in the background after the user provides the web application with camera and microphone permissions. This system takes video frames in real time, processes them in real time and generates alert tones at the time drowsiness pattern is detected. In case the model detects a yawn, a short 3 seconds alert sound is played to help get the attention of the driver. Notice of excessive drowsiness or constant eye closure of more than 5 seconds will however trigger a gradually increasing level of alarm playing, starting at a low volume and rising slowly to the point, after which the driver must respond or the eyes must be opened. Such a two-tier warning system reduces the level of unnecessarily disturbing warnings whilst maintaining prompt action towards rectifying the cause in case the fatigue is verified.

The web app will play alerts even when minimized or in the background. This feature is guarantees constant observation and alertness without interference with other activities on the screen.

4.6 Discussion

The findings distinctly show that CNN-YOLO-DeiT hybrid framework is a powerful and efficient method of detecting fatigue among drivers. The YOLOv11 is used to guarantee real-time localization and detection, CNN can effectively capture micro spatial patterns of blinking and yawning, and DeiT component represents temporal relationships in the long-term between frames. The combination can be used to both detect on a fine-grained scale and analyze long-sequences reliably, with the outcome being a higher degree of accuracy and a lower number of false positives.

The proposed approach has significant gains in detection accuracy, temporal cognition, and ability to respond to environmental changes in comparison to the traditional deep learning techniques. In addition, its web-based implementation means that the system is scalable, inexpensive, and non-intrusive, and therefore, an excellent fit in trucks and commercial vehicles that run in the Indian highways and conditions.

To sum up, the results of the experiment support the idea that CNN-YOLO-DeiT hybrid model is highly performing, has a real-time work, and can produce trustworthy alerts, which makes the model a promising AI-oriented solution to the problem of preventing fatigue-related road accidents among long-distance truck drivers.

V. Conclusion

The proposed CNN–YOLO–DeiT hybrid framework demonstrates a highly effective, real-time solution for detecting driver fatigue through spatial, temporal, and localization-based learning. By combining YOLOv11 for rapid region-of-interest detection, CNN for spatial feature extraction, and DeiT for temporal attention modeling, the system successfully overcomes the major limitations of earlier approaches such as poor temporal awareness, low robustness to lighting variations, and high latency. The model achieved a strong performance with a test accuracy of 94.6% and consistent robustness under challenging real-world conditions.

The proposed model integrates the powerful models: YOLOv11, CNN and DeiT. YOLOv11 is used for accurate and fast facial region localization, CNN is used for extracting intricate spatial features such as eye aspect ratio and mouth movements. And the DeiT model is used for capturing temporal dependencies and contextual information across sequential frames. These components allow the system to interpret not just isolated expressions but also behavioral patterns over time; the latter is an important differentiator for identifying real drowsiness rather than momentary blinks. The high accuracy of the model shows reliable performance. The model has a high test accuracy which shows reliable performance across lighting changes, face angles, and occlusions. It is deployed using a Flask-based web application that ensures real-time functionality without the need of specialized hardware. This makes it cost effective and easily accessible. The system uses two-tier alerts-a short audio tone for yawns and an escalating alarm for continuous sleepiness-to make sure the driver receives timely feedback to minimize the chances of accidents.

In summary, this study proposes an overall robust, scalable, and real-time fatigue detection solution suitable for integration with a smart transportation system. Overall, the proposed CNN-YOLO-DeiT hybrid approach not only improves the accuracy of detection but also promises real-world applicability toward meeting the urgent need for AI-powered road safety systems in the Indian logistics and transportation sectors. The study concludes that the integration of spatial, temporal, and attention-based models into a single hybrid framework is one important step further in the development of driver monitoring technologies and thus lays a solid foundation for future developments related to intelligent vehicle safety systems.

VI. Future Scope

The proposed hybrid model of CNN-YOLO-DeiT has shown immense potential in real-time driver drowsiness detection.

However, there is scope for future research on the subject. Future enhancements on this system would include increasing the precision and efficiency of the alert mechanism. The system at present issues sound alerts in case of detection of drowsiness or yawning; in future versions, this can be extended to automated emergency communications systems. This will include automatically placing a call or send an alert message to the driver's registered family members/transport company/nearby police station through integrated communication APIs if the driver remains unresponsive beyond a certain period of time. This will ensure timely corrective action like rescue in cases of emergencies, especially on highways or remote routes where human intervention may be delayed. Another enhancement would be the use of more sources of data such as heart rate, steering movement and voice monitoring to yield a more holistic picture of driver fatigue. Future work can also explore thermal imaging and night-vision integration for improving the accuracy of detection at night time or low-light driving conditions.

To further enhance real-world deployment, the system can be optimized for edge computing platforms, including the likes of NVIDIA Jetson Nano, Raspberry Pi, making offline operations with no constant need for internet connectivity possible. Integration with the IoT frameworks can enable real-time, cloud-based monitoring by fleet management companies and allow centralized supervision of many vehicles at once. Personalized driver profiles using machine learning will further tune the system to individual behavioural patterns, like the frequency of blinking or the structure of the face, hence reducing false alerts and increasing reliability. The proposed model is a substantial base to develop further AI-driven driver assistance systems. By integrating with emergency response, multi-modal sensing, and real-time cloud connectivity, this can expand into a comprehensive smart safety system that would go a long way in reducing road accidents and enhancing transportation safety in the country and beyond.

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